

Chapter 8

Verb phrase ellipsis in Likpakpaanl

Samuel Owoahene Acheampong

Goethe University, Frankfurt am Main

This paper discusses verb phrase ellipsis in Likpakpaanl, a Mabia (Gur) language belonging to the Niger-Congo family and predominantly spoken in the Eastern part of the Northern Region and the Oti Region of Ghana. I argue that VP ellipsis is licensed by the modals *ɲmàn* and *bàn*, which project separate modal phrases (ModP) in the clausal structure. The data drawn from the Licahbol dialect show that only the higher ModP position occupied by *ɲmàn* and *bàn* hosts the ellipsis [E]-feature which licenses VP ellipsis by deleting the phonological content of their complements. I show that even though both modals license VP ellipsis, *bàn* targets an entire CP for deletion, while *ɲmàn* deletes the VP and its internal arguments. Finally, I adduce evidence from Antecedent Contained Deletion (ACD) and Reconstruction effects and propose a base-generation analysis to account for the presence of the resumptive pronoun in the distribution of *bàn*, and also assume a PF-deletion approach for Likpakpaanl VP ellipsis.

1 Introduction

In natural language, we communicate through the use of words, which are strung together to convey our intended meaning. This meaning comes from the interlocutor's ability to create a meaningful link between sound and meaning. However, under ellipsis, the observed mapping of form-to-meaning is disrupted, resulting in a mismatch between form and meaning, although the meaning is not lost due to the omission of background information. This paper investigates the syntax of verb phrase ellipsis in Likpakpaanl and contributes to the ongoing effort to examine the nature of VP ellipsis licensing from a cross-linguistic perspective. There are few works on ellipsis in the Mabia languages, or African languages in general. The few works on ellipsis have come from Basàá (Bassong



Samuel Owoahene Acheampong. 2025. Verb phrase ellipsis in Likpakpaanl. In Akinbiyi Akinlabi, Sampson Korsah, Sharon Rose & Abdul-Razak Sulemana (eds.), *Cross-disciplinary approaches to Information Structure in Niger-Congo languages*, 181–213. Berlin: Language Science Press. DOI: 10.5281/zenodo.16875464 

2014), Gungbe (Lipták & Aboh 2013) and Nupe (Mendes & Kandybowicz 2023). Data from a less researched language like Likpakpaanl advances our understanding of the typological variation of VP ellipsis phenomena. The goal of this paper is to discuss VP ellipsis in Likpakpaanl and try to answer the following questions:

- i. What are the strategies for licensing VP ellipsis in Likpakpaanl?
- ii. Which functional element hosts the ellipsis feature that triggers VP ellipsis?
- iii. What evidence is there to support a PF-deletion account of Likpakpaanl VP ellipsis?

There are two main approaches in the study of ellipsis, viz the PF-deletion and null-Pro-form. This study adopts the former approach. The PF-Deletion approach, first proposed by Merchant (2001), argues that ellipsis involves an underlying syntactic structure that is present in Logical Form (LF) but deleted in Phonetic Form (PF). Under this analysis, the elided constituent in (1) contains the full syntactic structure, allowing it to be interpreted properly at LF, but the phonological content ‘eat pizza’ is deleted and, therefore, not pronounced (Merchant 2001, Frazier & Clifton 2005). PF-Deletion, as noted by Merchant (2001), applies post-syntactically at the interface between syntax and phonology and is driven by an ellipsis [E]-feature that triggers deletion as shown in the *strikeout* text in (1).

- (1) John ate pizza last night, and Mary did *eat pizza* too.

One advantage of the PF-Deletion approach is that it accounts for both the interpretability and silence inherent in ellipsis constructions while maintaining the presence of syntactic structure (Merchant 2004). Subsequent research has supported the PF-Deletion account through evidence from binding connectivity effects in ellipsis sites (Johnson 2001, Frazier & Clifton 2005, Aelbrecht & Harwood 2019).

Beyond this introduction, the rest of the paper is structured as follows: Section 2 explains the notion of ellipsis in general and verb phrase ellipsis in particular, while Section 3 provides a background to the language and the causal structure. Section 4 introduces the various modal auxiliaries and their interaction with the tense and aspectual systems of the language and lays the basis for discussing VP ellipsis in the language. Section 5 examines the various licensing conditions for VP ellipsis and shows that not all modals can license VP ellipsis except *ɲmàn* and *bàn*, while Section 6 presents the evidence in support of a

movement plus PF deletion. Finally, Section 7 summarises the paper's findings and conclusions.

2 The notion of ellipsis

Ellipsis refers to a linguistic phenomenon where some elements of a sentence are omitted, but we are still able to interpret the meaning based on the context of the sentence (Ross 1967, Merchant 2019). Ellipsis has been identified as a cross-linguistic phenomenon common in most languages. However, the language-internal properties of each language play a crucial role in determining the licensing conditions of ellipsis. For instance, Dagnac (2010) observed that in French and Spanish, it is modal verbs that license a type of VP ellipsis, which they term “Modal complement ellipsis” (MCE) (Aelbrecht 2010). In the examples in (2), the crossed-out materials represent unpronounced syntactic constituents (the ellipsis site), which, though present in the syntax, are not overt at LF (Johnson 2001, Merchant 2001, Aelbrecht 2010, Dagnac 2010). From the literature, different ellipsis phenomena, such as nominal ellipsis, gapping, sluicing, and verb phrase ellipsis, are derived from syntactic movement followed by the deletion of the phonological form (PF) part of the clause (Merchant 2001, Aelbrecht 2010, van Craenenbroeck 2017). Verb Phrase ellipsis (VPE), which is the focus of this paper, describes a linguistic phenomenon where the verb phrase, together with its internal arguments, are elided if they meet the licensing and identification conditions proposed by Merchant (2001) and Lobeck (1995). Merchant (2001) explains licensing as a locality constraint on syntactic heads that determine the structure in a particular language that can be omitted under ellipsis, while identification (syntactic and semantic) refers to the process of recovering missing information that is expected to be present in overt syntax. Under the PF-deletion approach to VP ellipsis, the “deleted” predicate in (2) gets a full semantic interpretation because it has an appropriate antecedent in the first conjunct clause. The predicates in the second conjunct of (2a) and (2b) are not overtly expressed, yet the sentences are interpreted as ‘Anthony will eat the plantain’ and ‘I could buy Susan the oranges’, respectively.

- (2) a. The children will eat the plantain, and Anthony will [~~eat the plantain~~] too.
 b. Adelina couldn't buy Susan the oranges, but I could [~~buy Susan the oranges~~].

Johnson (2001), Merchant (2001) and Aelbrecht & Harwood (2015) note that research into VP ellipsis has revolved around trying to examine the syntactic environments in which it is licensed, investigating the structural relationship between the elided verb phrase and its antecedent, and exploring how the meaning at the ellipsis site is recovered. Johnson (2001) and Aelbrecht & Harwood (2015) point out that finite auxiliaries license VP ellipsis in English. Thus, the example in (3) shows that the finite auxiliaries *could* and *had* cannot be “deleted” because they license VP ellipsis in English. However, lexical verbs, whether finite or non-finite, need to be deleted in VP ellipsis as illustrated in (3a). If a sentence does not have a finite auxiliary to license the ellipsis in English, a dummy auxiliary ‘do’ is inserted in the structure, as in (3d); otherwise, the resulting sentence will be ungrammatical. Mutual entailment between an antecedent clause and an elided clause is necessary for the felicitous interpretation of verb phrase ellipsis (VPE) (Dalrymple et al. 1991). Dalrymple et al. (1991) note that in VPE like (3a) and (3b), the interpretation is possible because the antecedent clause *I thought the auxiliary hadn’t disappeared* and the elided clause *it had* stand in a mutual entailment relationship, such that each clause entails the other. This creates the parallelism required for resolving the missing verb ‘disappeared.’ Consequently, in (3c), the VPE interpretation fails, indicating that mutual entailment between the antecedent and elided clauses is necessary for licensing VPEs and arriving at the intended interpretation (Dalrymple et al. 1991, Hardt 1999).

- (3) a. An elephant can’t fly, but maybe a rhino *(could) [~~fly~~].
b. I thought the auxiliary hadn’t disappeared, but it *(had) [~~disappeared~~].
c. *The chicken didn’t put the tuna on the table, but the penguin put [~~the tuna on the table~~].
d. The chicken didn’t put the tuna on the table, but the penguin did [~~put the tuna on the table~~].

(Aelbrecht & Harwood 2015: 63).

As a general principle, elements that undergo ellipsis must be subject to the recoverability condition, which requires that whatever is elided should be able to establish a mutually entailed interpretation between the antecedent and the ellipsis site. Thus, elements at the ellipsis site must be retrievable via a salient linguistic antecedent (Johnson 2001, Merchant 2001, van Craenenbroeck 2017). Fiengo & May (1994) view the identity relation between the ellipsis site and the antecedent as a pure syntactic identity and propose a “Strict Isomorphism” condition, which requires elided elements to have a syntactically identical antecedent.

3 Background to Likpakpaanl and the clausal structure

This sub-section briefly explains the properties of Likpakpaanl to provide a background for understanding the discussion of VP ellipsis licensing. Likpakpaanl (also known in the literature as Konkomba) belongs to the Gurma subgroup of the Oti-Volta branch of North Central Mabia¹ (Manessy 1971, Naden 1989). The people refer to themselves as *Bikpakpaam*, their language as *Likpakpaanl* and their land as *Kikpakpaanl*. The language is predominantly spoken in the Eastern Corridor of the Northern region of Ghana as well as in the North-East, Oti and Ahafo regions. Likpakpaanl is also spoken in the Western part of the Republic of Togo, with a population of 112,000 and over 1,187,000 speakers in Ghana (Joshua Project 2022, World Population Review 2023).

Likpakpaanl has a system of 15 noun classes based predominantly on class affixes, which also bear number semantics of number agreement (Tait & Strevens 1954, Winkelmann 2012, Bisilki & Akpanglo-Nartey 2017). Noun class affixes are used to distinguish singular and plural nouns. The affixes also play a central role in marking agreement patterns in the internal structure of the DP, such as demonstratives and possessives. Structurally, Likpakpaanl, like other Mabia languages, is an SVO language such that a canonical simple sentence can have the pattern SVO, SV or SVA, depending on whether the verb is transitive or intransitive, as illustrated in (4a), (4b) and (4c), respectively.²

- (4) a. Tàmànjà nà̀n tər̀ ù-bò gbà̀n. (SVO)
 T. PST help 1-child DEF
 ‘Tamanja helped the child.’
- b. Wumbei là̀. (SV)
 W. laugh.PFV
 ‘Wumbei has laughed.’
- c. Naja nà̀n chù̀n sù̀-sù̀. (SVA)
 N. PST walk slow-slow
 ‘Naja walked slowly.’

In ditransitive constructions, the verb is immediately followed by the indirect object before the direct object and adverbials, as illustrated in (5a). Sentence (5b)

¹I use the term ‘Mabia’ in the sense of Bodomo (2020) to refer to over 80 languages spoken across the West African sub-region. The Mabia languages, numbering about 70 languages, are also spoken in south-eastern Mali, Burkina Faso, northern Benin, northwestern Nigeria and northern Ivory Coast.

²Odd numbers represent singular noun affixes, while even numbers encode plural noun classes

illustrates a serial verb construction, where the introduction of the verb *dì* selects the direct object, which precedes the indirect object, as its internal argument.

- (5) a. Adelínà gà tì Ama lí-nùù-l gbààn dín.
 A. FUT give A. 7-yam-7 DEF today
 ‘Adelina will give Ama the yam today.’
 b. Kànjo gà dì lí-nùù-l gbààn tì Ama.
 K. FUT take 7-yam-7 DEF give A.
 ‘Kànjo will give the yam to Ama.’

Despite the SVO clause structure, information-structural-related constructions can trigger the extraction of these elements out of their canonical positions and moved to the sentence-initial position, resulting in some cases to focus (6b) and topic (6c) interpretations.

- (6) a. Ponaà dàà ì-ṇùò. (Canonical)
 P. buy.PFV 6-goat
 ‘Ponaà has bought goats.’
 b. Ì-ṇùò_i lé Ponaà dàà t_i. (Focus)
 6-goat FOC P. buy.PFV
 ‘Ponaà has bought GOATS.’
 c. Ì-ṇùò gbààn_i mà-kàn Ponaà dàà t_i. (Topic)
 6-goat DEF TOP P. buy.PFV
 ‘As for the GOATS, Ponaà bought them.’

It is also worth noting that Likpakpaanl verbs do not inflect for tense, aspect, person, or number. Therefore, tense, aspect, and mood are marked by independent preverbal particles in the language. In Likpakpaanl, the present tense is not morphologically marked; the past tense is marked by preverbal particles that categorise into immediate, external, and remote past. The immediate past tense or hodiernal, marked by the particle *ba*, describes events or actions that occurred the same day or shortly before the current moment of speaking (7a). There is the hesternal past tense or “yesterday past” (7b), which refers to events that took place exactly a day before the speech time. Events that happened more than three days, a week, to years before the present moment are marked by the remote past particle *nàn* as in (7c) (glossed as PST), while the future tense is illustrated in (7d).

- (7) a. Mbeijom bà jìn sàkolà.
 M. HOD.PST eat 3.fufu
 ‘Mbeijom ate fufu (within the day).’

- b. Mbeijom fè jìn sàkòlà.
 M. HEST.PST eat 3.fufu
 ‘Mbeijom ate fufu (yesterday).’
- c. Mbeijom nàn jìn sàkòlà.
 M. PST eat 3.fufu
 ‘Mbeijom ate fufu (some time ago).’
- d. Mbeijom gà jìn sàkòlà.
 M. FUT eat 3.fufu
 ‘Mbeijom will eat fufu.’

The perfective aspect in Likpakpaanl, like many other Mabilia languages, is unmarked and a bare verb such as *màà* in (8a) receives a perfective interpretation, signalling that an event or action has been completed. The imperfective aspect is morphologically expressed by a preverbal particle *bì* as in (8b) and demonstrates that the act of “building” is still ongoing at the time of the utterance (see also, Bodomo 1997, Issah 2013, Abubakari 2019).

- (8) a. Kando màà kì-dìi-k.
 K. build.PFV 11-room-11
 ‘Kando has built a room.’
- b. Kando bì màà kì-dìi-k.
 K. IPFV build 11-room-11
 ‘Kando is building a room.’

The existence of such preverbal particles is a key property of the clause structure of the Mabilia languages like Dagaare (Bodomo 1997, 2020), Dagbani (Issah 2013, 2020), Kusaal (Abubakari 2018, 2019), and Gurene (Kropp Dakubu 2003, Atintono 2013) among others.

4 Interaction between modals, tense and aspect in Likpakpaanl

This sub-section provides a background to modal construction in Likpakpaanl, the basis upon which the discussion of VP ellipsis will be centred. The modal system of verbs in Likpakpaanl is denoted by particles that express root and epistemic modality. Kratzer (1981) and Nuyts (2001), among others, have argued that modals constitute quantifiers that create possible “worlds” whose restriction is contextually provided. She identifies two broad classes of modals: epistemic

modals and root modals. Kratzer (1981) notes that while the former express a speaker's assessment of the probability of a proposition being true, focusing on the speaker's knowledge and opinions about the world, the latter, conversely, convey notions of obligation, permission, or necessity and thus indicate whether an action is required, allowed, or forbidden (Hacquard 2006, 2010). According to Kratzer (1981), there are two broad classes of modals: epistemic and root modals; the former express a speaker's assessment of the probability of a proposition being true, focusing on the speaker's knowledge, confidence or lack of it in the truth of a proposition. Examples include *might*, *could*, *may*, *must*, *probably*, and *likely*. Root modals, conversely, convey notions of obligation, permission, or necessity and thus indicate whether an action is required, allowed, or forbidden (Nuyts 2001, Hacquard 2006, 2009).

4.1 The syntax of Likpakpaanl modal verbs

Likpakpaanl modal elements like *ɲmàn* 'can', *bàn* 'want' and *nibààkàn*³ 'maybe' are used to encode ability, possibility, and probability, respectively. A root modal like *ɲmàn* in the examples in (9a) and (9b) expresses ability, indicating that Takai and Mpopiin' are able to pluck the mango and weed the farm, respectively.

- (9) a. Takal nàn ɲmàn kìr mǎngù gbààn.
 T. PST can pluck mango DEF
 'Takal was able to pluck the mango.'

³Adopting Hoyer (1997) and Nuyts (2001)'s criteria for modal adverbs, I analyse *nibààkàn* as a modal adverb based on its syntactic patterning. Examples (i.a)-(i.c) demonstrate *nibààkàn* can appear in a clause's initial, medial, or final position without any change in its meaning.

- (i) a. Nibààkàn Jàbàb gà dàn dìn.
 maybe J. FUT come today
 'Maybe/Probably, Jabab will come today.'
 b. Jàbàb gà dàn dìn, nibààkàn.
 J. FUT come today maybe
 'Jabab will come today, maybe/probably.'
 c. Jàbàb gà dàn, nibààkàn dìn.
 J. FUT come maybe today
 'Jabab will maybe/probably come today.'

This positional flexibility indicates that it syntactically takes sentential scope and modifies the truth value of the entire proposition rather than a single constituent. Semantically, it expresses epistemic modality by qualifying the likelihood of the proposition 'Jabab will come' being true.

- b. Mpopiin nàn ɲmàn kùùr kí-sàà-k gbààn.
 M. PST can weed 11-farm-11 DEF
 ‘Mpopiin was able to weed the farm.’

In its distribution, another modal *bàn* can optionally occur with the complementiser *kè* to express desires, preferences, intentions, or requests. For instance, the use of *bàn* in (10a) demonstrates the intentional willingness of *Powon* to go to school while (10b) expresses *Kunji*’s desire to eat groundnuts. Example (10c) also illustrates that the same modal can be used to make a polite request.

- (10) a. Powon_i bàn (kè) ù_i bùén sikùù.
 P. want that 3SG go 3.school
 ‘Powon wants to attend school.’
 b. Kunji_i bàn (kè) ù_i ɲmɔ simà.
 K. want that 3SG eat 3.groundnuts
 ‘Kunji wants to eat groundnuts.’
 c. M bàn (kè) à tér mi.
 1SG want that 2SG help me
 ‘I want you to help me.’

A major observation that can be made from the distribution of *bàn* is that it selects a CP complement introduced by the optional complementiser *that*. Thus, the sentence in (10a) translates literally as ‘Powon_i wants that he_i goes to school.’ Such sentences are ambiguous since the pronoun could also refer to someone other than ‘Powon’, and the only way to disambiguate it is by using the appropriate co-indexation or situating it within the right context.

In Likpakpaanl, interpretations of obligation and necessity arise when the copular *yè* ‘be’ co-occurs with the complementiser *kè* ‘that’. Thus, the bi-partite modal *yè* *kè* denotes an objective necessity to indicate that something has to happen. The interpretation of *yè* *kè* in (11) means that a ‘child’ needs to obligatorily submit to the authority of elders as a socio-cultural requirement, as far as the African society is concerned.

- (11) Û-bo yè kè ù pàk bí-nínkpíí-b.
 1-child COP that 3SG respect 2-elder-2
 ‘A child must respect elders.’

Given different contexts, the expression *yè* *kè* can elicit different interpretations. For example, in a context where there are dark clouds, a sentence like (12a)

is uttered as an inference of a high probability, based on observable evidence, that rain is likely to fall. On the contrary, the *yè kè* in (12b) points to the fact that Binlù has to fulfil an obligation and ensure that the meat is fried.

- (12) a. Nì yè kè ù-tààl nù.
it COP that 1-rain fall
'It must rain.'
b. Nì yè kè Bìnlù kìn tì-nàn gbààn.
it COP that B. fry.PFV 14-meat DEF
'Binlu must fry the meat.'

It is also important to note that a construction with only the copular verb, as presented in (13a), yields a cleft-focus construction and requires the focus particle for the sentence to be grammatical. Example (13b) is, therefore, ungrammatical because of the absence of the complementiser *kè* to convey the desired meaning.

- (13) a. Nì yè Bìnlù lé kìn tì-nàn gbààn.
it COP B. FOC fry.PFV 14-meat DEF
'BINLU fried the meat/ It is BINLU who fried the meat.'
b. *Nì yè Bìnlù kìn tì-nàn gbààn.
it COP B. fry.PFV 14-meat DEF
Intended: 'Binlu must fry the meat.'

The use of bi-partite markers to express modality is not new in the literature of African languages. Many Niger-Congo languages, including Lingala (Heine 1991), Ewe (Ameka 2008), and Igbo (Uchekchukwu 2008), among others, use bipartite markers comprising two distinct elements to convey modal interpretations. According to Ameka (2008), in Ewe, an obligation is expressed when the third person singular pronoun *é-* is combined with the expletive verb and followed by the complement clause. In (14), the 3SG-pronoun, which is the subject, is followed by a verb and the complementiser *bé*, which introduces the complement clause to express the obligation.

- (14) Ewe (Ameka 2008: 147)
É-dze bé nà-vá kpɔ-m.
3SG-fit that 2SG:SUBJV-VENT see-1SG
'You must come to see me.'

Having looked at root modals, I also briefly examine the distribution and use of some epistemic modals. These modals indicate a speaker's degree of certainty, familiarity with a proposition or judgment about the likely occurrence of a proposition rather than its truth value. In (15), the epistemic modal expresses the general

possibility that Pando will buy a canoe. *Nibààkàn*, meaning ‘probably/maybe,’ belongs to a group of modals which have been referred to in the literature as modal adverbs (Nuyts 1993, Hoyer 1997, Cinque 1999, Nuyts 2001). Example (15a) demonstrates that the epistemic adverb can also appear in the verbal domain of the clause without significant differences in the semantic interpretation. It does not immediately come before the verb as a modal adverb but can occur in clause-initial (15a), medial (15b) or final position (15c). In (15c), the modal adverb occurs between the raised subject in the specifier of a topic phrase (Spec-Top) and the tense marker. The base position of the moved subject is replaced with a resumptive pronoun (see also the Italian example in (16b)).

- (15) a. *Nibààkàn*, Pando gà dàà bù-ɲɔb.
 maybe P. FUT buy 9-canoe
 ‘Maybe/probably, Pando will buy a canoe.’
 b. Pando gà dàà bù-ɲɔb, *nibààkàn*.
 P. FUT buy 9-canoe maybe
 ‘Pando will probably buy a canoe.’
 c. Pando_i mà-kán, *nibààkàn*, ú_i gà dàà bù-ɲɔb.
 P. TOP maybe RP FUT buy 9-canoe
 ‘As for Pando, he will buy a canoe.’

The Likpakpaanl modal adverb shares some similar syntactic behaviour with its Italian counterpart *probabilmente* ‘probably’. According to Giorgi (2013) *probabilmente* is able to semantically scope over an entire sentence whether it precedes or follows the subject without resulting in a change of meaning as in (16a). Giorgi (2013) predicts that modal adverbs can either occur above the tense phrase (TP) in the clause-initial position or in the base-generated position as the complement of the verb.

- (16) Italian (Giorgi 2013: 99)
 a. *Probabilmente* Gianni vincerà la gara.
 probably G. win the race
 ‘Probably Gianni will win the race.’
 b. Gianni *probabilmente* vincerà la gara.
 G. probably win the race
 ‘Probably Gianni will win the race.’

Furthermore, the modal *li* translated literary as ‘should’ can also be used to express either epistemic or root modality. In (17a), *li* conveys necessity or obligation by indicating that *fathers* are required, according to rules, expectations, or

moral standards, to take good care of their children. In example (17a), however, in a context where someone sees Wongol entering a compound and people follow up and do not find him in the yard, then the sentence in (17b) is uttered to express a degree of possibility that once Wongol was in the yard, then the belief is that he is in one of the rooms. The examples above show that the context determines whether *lì* functions as a root or epistemic modal.

- (17) a. Tè-tiib lì lìk bì àà-bim pù.
 father-2 should look 3SG POSS-child LOC
 ‘Fathers should look after their children.’
 b. Wongol gà lì kòò kì-dù-k nì.
 w. FUT should enter 11-room-11 in
 ‘Wongol should be in the room.’

The data presented so far have demonstrated that some modal verbs can convey both root and epistemic modality, thus expressing both the underlying necessity of actions or events and the speaker’s degree of certainty. The fact that the modals can be used to express root and epistemic modality in (17a) and (17b) is not novel since the same observation has been made in many different languages. For instance, Hacquard (2016) contends that the modal ‘must’ can express either root or epistemic modality in English. The sentence in (18a) expresses epistemic modality because the speaker believes that if John is not in his office, then he will certainly be at home. The same modal ‘must’ in the (18b) denotes an obligation for Dickson to pay the fine for wrong parking. He has no choice.

- (18) English (Hacquard 2016: 45)
 a. John is not in his office. He must be home.
 b. Dickson parked illegally. He must pay a fine.

Additionally, Butler (2003), highlighting the difference between root and epistemic modals in Yoruba, notes that the modal *gbòdó* in (19) could have a dual interpretation suggesting a strong conviction for a young person to marry or an obligation for them to do so.

- (19) Yoruba (Butler 2003: 972)
 Ó gbòdó ní iyàwò.
 he must have bride
 ‘He must be married.’ (root/epistemic)

As a result of these subtle interpretational differences in modal verbs, many authors have claimed that the distinction can be accounted for syntactically or lexically even though other scholars like Kratzer (1981), Cinque (1999), Butler (2003), Hacquard (2016) among others, also assert it is semantically or pragmatically determined.

4.2 Interaction between modal, tense and aspect

From this modal construction overview, I will examine the interaction between modals and the aspectual system in Likpakpaanl. The interaction between tense, aspect and modal in Likpakpaanl elicits different interpretations. In (20a), we have a default perfective verb co-occurring with the modal *ɲmàn* to express the past perfective aspect of Bilikuni possessing the ability to fix the bicycle. In (20b), the auxiliary verb ‘will’ is used as a root modal to demonstrate the capability of Bilikuni to repair the bicycle in the future. In (20c), tense and aspect interact with two modals to suggest a moderate degree of certainty that Bilikuni could be riding a bicycle, but it does not imply necessity or obligation.

- (20) a. Bilikuni ɲmàn tòòr chééchè gbààn.
 B. can repair.PFV 3.bicycle DEF
 ‘Bilikuni has been able to repair the bicycle.’
- b. Bilikuni gà ɲmàn tòòr chééchè gbààn.
 B. FUT can repair.PFV 3.bicycle DEF
 ‘Bilikuni will be able to repair the bicycle.’
- c. Bilikuni gà ɲmàn lì bì tòòr chééchè gbààn.
 B. FUT can should IPFV repair.PFV 3.bicycle DEF
 ‘Bilikuni could be repairing the bicycle.’
- d. Bilikuni ààn ɲmàn lì bì tòòr chééchè gbààn.
 B. NEG.FUT can should IPFV repair.PFV 3.bicycle DEF
 ‘Bilikuni could not be repairing the bicycle.’

Moreover, in the sentences in (21), using different tense and aspectual markers in the presence of the same modal verb serves different modal interpretations related to time and certainty. The use of *lì* in (21a) expresses a sense of past necessity where Noah was expected to buy a canoe. The use of *lì* in (21b) indicates a future possibility or likelihood of Noah acquiring a canoe. Example (21b), therefore, suggests that there is a good chance that Noah will buy a canoe in the future, but does not necessarily imply a necessity as in (21a).

- (21) a. Noah bà lì dàà bù-ḡɔb gbààn.
 N. HOD.PST should buy 9-canoe DEF
 ‘Noah should have bought the canoe.’
 b. Noah gà lì bì dàà bù-ḡɔb gbààn.
 N. FUT should IPFV buy 9-canoe DEF
 ‘Noah would be buying the canoe.’

Descriptively, the examples from (20) and (21) show that in the interaction between tense, aspect and modals in Likpakpaanl, the modal phrase occurs below the TP but above aspect and the VP. Thus, in (22a), the modal *ḡmàn* and *lì* project below TP but above the aspectual phrase and the vP. The ungrammaticality of (22c) occurs because the aspectual phrase precedes the modal. The data further indicates root modals can co-occur, and when they do, *ḡmàn* referring to ‘ability’ must precede, *lì* as captured in the ungrammaticality of (22c).

- (22) a. [TP Binlu gà [ModP ḡmàn [ModP lì [IPFV bì [vP_i kìr [t_i
 B. FUT can should IPFV pluck
 màngù.]
 2.mango
 ‘Binlu could be plucking a mango.’
 b. [TP Binlu ààn [ModP ḡmàn [ModP lì [IPFV bì [vP_i kìr [t_i
 B. NEG.FUT can should IPFV pluck
 màngù.]
 2.mango
 ‘Binlu would not be plucking a mango.’
 c. *[TP Binlu gà [ModP lì [IPFV bì [ModP ḡmàn [vP_i kìr [t_i
 B. FUT should IPFV can pluck
 màngù.]
 2.mango
 Intended: ‘Binlu could be plucking a mango.’

The data on the interaction between the various modals in Likpakpaanl brings into question the hierarchy of appearance. The relative ordering of modals as functional categories within the causal architecture has been the subject of considerable research, and the general consensus is that epistemic modals are higher than roots. Following Cinque (1999), I assume that Likpakpaanl modal verbs are heads of functional projections of a Modal Phrase (ModP), which project above aspect and below the tense Phrase (TP). The various modals such as *ḡmàn*, *lì*, *bàn*, etc., constitute different sub-categories of modals in the hierarchy of root modals in Likpakpaanl as proposed by Cinque (1999).

5 Derivation of Likpakpaanl VP ellipsis

In the previous section, we examined the distribution of modal auxiliaries, and the discussion in this section proceeds to discuss how modals license the derivation of VPE in Likpakpaanl. Even though Section 4.1 showed the presence of different modals, I show that among the various modals, it is only *ɲmàn* and *bàn* that are able to license VP ellipsis in the language. I also argue that when these two modals co-occur, the functionally higher modal *ɲmàn*, which bears the [E]-feature, licenses the deletion. Likpakpaanl illustrates a type of VP ellipsis where the verb phrase and its internal argument in the complement of a modal auxiliary are deleted. The examples will show that the modal can elide its complements and still have the same interpretation in relation to an antecedent clause. Such phenomena have been found in Dutch (Aelbrecht 2010), Italian (Lobeck 1995) and French Dagnac (2010) and they refer to this type of ellipsis as *Modal Complement Ellipsis* (MCE).⁴

5.1 The syntax of *ɲmàn*

The first modal auxiliary I consider is the root modal *ɲmàn*. In (23a), it elides its VP complement, and this is possible because the elided verb phrase *read the book* has an appropriate antecedent in the first conjunct. The elided constituent is semantically understood to mean *Kunji can't read the book*. As predicted by the identify and e-GIVENness condition, (23b) is disallowed under ellipsis since the deleted VP (the struck-out element in the ellipsis site) is new in the discourse and therefore fails to satisfy the e-GIVENness condition, which stipulates that an element, a VP, can be elided if it has an appropriate antecedent and has a mutually entailing relationship with the antecedent. Thus, only previously mentioned elements in a discourse can undergo ellipsis (see Schwarzschild 1999, Merchant 2001).

- (23) a. Ama bà ɲmàn kàrn kì-gbàŋ gbààn àmàà Kùnjí àà bà
 A. HOD.PST can read 11-book DEF but K. NEG HOD.PST
 ɲmàn [kàrn kì-gbàŋ gbààn.] .
 can [read 11-book DEF]
 ‘Ama can read the book, but Kunji can’t.’
 b. *Ama bà ɲmàn kàrn kì-gbàŋ gbààn àmàà Kùnjí àà bà
 A. HOD.PST can read 11-book DEF but K. NEG HOD.PST
 ɲmàn [kìrì ì-jàn gbààn.]
 can [fry 6-fish DEF]

⁴I still use the conventional term “Verb phrase ellipsis” to refer to the phenomenon in Likpakpaanl

Apart from occurring in local clauses, *ɲmàn* is also able to license VP ellipsis in non-local clauses such as (24a) where it elides the entire VP complement inside the embedded clause. Thus, the unacceptability of (24b) can be explained in terms of its violation of the e-GIVENness and identity condition on ellipsis.

- (24) a. Nèinà bì sà̀n sà̀kòlà lè Olàmbà; mù dàk kè ù; gà ɲmàn
 N. IPFV pound fufu and o. too think that 3SG FUT can
 [sà̀n sà̀kòlà.]
 [pound fufu]
 ‘Neina is pounding fufu, and Olamba thinks he will be able to ____.’
- b. *Nèinà bì sà̀n sà̀kòlà lè Olàmbà; mù dàk kè ù; gà ɲmàn
 N. IPFV pound fufu and o. too think that 3SG FUT can
 [yèí lì-nùù-l.]
 [peel 7-yam-7]
 Intended: ‘Neina is pounding fufu, and Olamba thinks he can peel
 yam-too.’

Ungrammatical sentences like (24b) demonstrate that syntactic identity is essential in licensing VP-ellipsis because only constituents that satisfy the syntactic identity condition can be elided. Merchant (2013), however, notes that there are instances where VP ellipsis permits voice mismatches between elements at the ellipsis site and the antecedent. For example, in (25), the verb phrase in the antecedent clause is in the active voice, but the elided VP occurs in the passive structure. Despite the fact that (25) violates the syntactic identity condition on ellipsis since the passive phrase is elided based on the active antecedent, the sentence is nevertheless grammatical.

- (25) The janitor must remove the trash whenever it is apparent that it should
 be removed. (Merchant 2013: 78).

The e-GIVENness condition requires a mutual entailment relationship between the ellipsis site and the antecedent as illustrated in the example in (26), adapted from Merchant (2001) where the most appropriate and accepted interpretation is (26a), and not (26b). The difference in interpretation between the two examples is because the content of the antecedent verb phrase in (26b) ‘...insult Francis’ does not express a logical relationship or establish the true condition presented in (26a), ‘...call Francis an idiot’ has a semantically entailed interpretation with the elided VP.

- (26) a. Adelina called Francis an idiot after Bianca did [~~call Francis an idiot~~].
 b. ≠Abby called Chunk an idiot after Ben did [~~insult Francis~~].

Thus, it is possible to elide the VP in (26a) because it is e-GIVEN, and its interpretation is entailed in the antecedent. The sentence in (26b) is infelicitous because it lacks an appropriate antecedent. Thus, it cannot be elided because it violates the syntactic and semantic identity condition of ellipsis because Bianca can insult Francis without necessarily ‘calling him an idiot.’ Therefore, it falsifies the statement in the antecedent VP. It is, however, important to state that the antecedent is not always a linguistic element because an elided constituent can also find an antecedent in the non-linguistic context (Merchant 2004, Biloa & Bassong 2020). There is also evidence that other syntactic elements, such as temporal adverbs and adjectives, can also be contained in the E-site in VP ellipsis in Likpakpaanl. For instance, in (27a), the VP and its internal arguments, the DP *woman*, and the temporal adverb are all elided, while in (27b), the modal deletes the verb *dàà* ‘buy’ together with the DP as well as the numeral and qualitative adjective modifiers.

- (27) a. Bileṇàn lèn kè Ntaadiù; gà yòòr ù-píí lí-bín-l lí-mìnà, àmàà tí
 B. say that N. FUT take 1-wife 7-year-7 7.DEM but 1PL
 mómòk nyì kè ù; ààn ṇmàn [yòòr ù-píí lí-bín-l lí-mìnà.]
 all know that 3SG NEG.FUT can [take 1-wife 7-year-7 7.DEM]
 ‘Bileṇàn said that Ntaadi would marry this year, but we all know that
 he can’t [~~marry this year~~].’
 b. Takai dàà ṇì-nààtāk pùàn ṇì-lèè tìi ù-nà lè Nàntòb; mù
 T. buy 8-sandal new 8-two give 3SG-mother and N. too
 dàk kè ù; gà ṇmàn [dàà ṇì-nààtāk pùàn ṇì-lèè tìi
 think that 3SG FUT can [buy 8-sandal new 8-two give
 ù-nà.]
 3SG-mother]
 ‘Takai bought two new sandals for her mother, and Nantob thinks
 (that) she can [~~buy two new sandals for her mother~~], too.’

5.2 The syntax of *bàn*

The data examined so far has looked at *ṇmàn* as a licenser for VP ellipsis in Likpakpaanl. This sub-section discusses how *bàn* licenses the deletion of the PF-content of its complement. The semantic interpretation of the elided constituent is due to the fact that its meaning can be inferred and “recovered” from the antecedent clause in the first conjunct because of the mutual entailment between the unpronounced constituent in the ellipsis site and the antecedent.

The data in (28a) illustrate the fact that ellipsis is licensed by only *bàn* where the CP complement *that he slaughter a pig* in the second conjunct undergoes dele-

tion. The ungrammaticality of (28b) and (28c), on the other hand, demonstrates that VP ellipsis in Likpakpaanl is blocked if the licensing head is a complementiser or a resumptive pronoun supporting Merchant's (2001) assertion about the selectional restrictions on lexical items in a particular language that possess the required [E]-feature to license VP ellipsis and ellipsis in general.

- (28) a. Dawòn bàn (ké) ù kər í-gbèèr lè Naadi mù bàn [(ké) ù
D. want that RP slaughter 6-pig and N. too want [that RP
kər í-gbèèr.]
slaughter 6-pig]
'Dawon wants to slaughter a pig, and Naadi also wants to.'
- b. *Dawòn bàn (ké) ù kər í-gbèèr lè Naadi mù bàn (ké) [ù
D. want that RP slaughter 6-pig and N. too want that [RP
kər í-gbèèr.]
slaughter 6-pig]
Intended: 'Dawon wants to slaughter a pig, and Naadi also wants to.'
- c. *Waatan bàn (ké) ù sànbànché béé ù àà bàn (ké) ù
W. want that RP pound 2.cassava or 3SG NEG want that RP
[sànbànché.]
[pound 2.cassava]
Intended: 'Does Waatan want to pound cassava or she doesn't want to.'

The distribution of *bàn* in the examples above shows the presence of an optional complementiser *ké* 'that' and an obligatory resumptive pronoun. Thus, unlike *ɲmàn*, which targets a VP for deletion, *bàn* selects a CP as its complement. I argue that even though both *bàn* and *ɲmàn* target their complements for deletion, they vary in the size of the ellipsis site they elide. Thus, while the former targets an entire complementiser phrase (CP) for deletion, the latter elides just its VP complement and its internal arguments.

The critical question is how to account for the occurrence of the resumptive pronoun in the syntax of *bàn*. The data on the distribution of *bàn* show a resumptive pronoun (RP) in the embedded clause, and I contend that the presence of the RP suggests that the subject in the matrix clause is base-generated in Spec-TP in the left periphery of the clause and binds the RP. The motivation for such an assumption comes from island and reconstruction effects in Likpakpaanl. The specifier of the CP has been argued to constitute an "escape hatch" for elements that are undergoing movement out of embedded clauses (Chomsky 2000). Following that assertion, one will expect (29b) to be ungrammatical if the RP is a

remnant of the moved subject. This is because the intermediate landing site of the moved DP is already occupied by another DP island and, therefore, blocking any movement out of it. Thus, (29b) is grammatical because the subject, Talido, as the antecedent c-commands and therefore binds the resumptive pronoun in its base-generated position (Chomsky 1980, 1981, Reinhart & Reuland 1993).

- (29) a. Talido nàn bàn ké ù dì i-líl gbààn dàà nì-tùùn.
 T. PST want that RP take 6-money DEF buy 8-beans
 ‘Talido wanted the money to buy beans.’
- b. Talido nàn bàn [DP i-líl gbààn_t [CP ké [TP ù dì t_i dàà
 T. PST want 6-money DEF that RP take buy
 nì-tùùn.]]]
 8-beans
 ‘Talido wanted the money to buy beans.’

Using reconstruction as a diagnostic for movement, I propose a base-generation account in examples such as (30b) where the RP is unable to reconstruct because it is not c-commanded at LF by the antecedent that binds it and, therefore, cannot receive a bound reading. Thus, if reconstruction is understood as a property of chains created by movement as argued by Chomsky (1980), then the data suggest there is no movement involved in (30b) but that the RP is bound by Kotin.

- (30) a. Kotin_i gà bàn ké ù_i tèr ù-bà_i.
 K. FUT want that RP help 3SG-self
 ‘Kotin will want to help himself.’
- b. *Ù-bà_i lé Kotin_i gà bàn ké ù_i tèr t_i.
 3SG-self FOC K. FUT want that RP help
 Intended: ‘Kotin will want to help himself.’

5.3 VP ellipsis licensing restrictions in Likpakpaanl

The literature on ellipsis has shown that cross-linguistically, the [E]-feature is syntactically restricted, and as a result, not every lexical item can be a licenser for ellipsis. Having focused on the modals *ɲmàn* and *bàn*, the rest of this section looks at other modal verbs like *yè kè* ‘must’, *nibààkàn* ‘maybe’ and *lì* ‘should’ as well as tense, aspect and negation particles and show that they cannot license VP ellipsis in the language. The ungrammatical sentences in (31a) and (31b) involving the modal *yè kè* lends credence to that conclusion.

- (31) a. *Sukpen yè kè ù pén kí-dìi-k gbààn àmàà Mpopiin àà yè
 s. COP that RP roof 11-room-11 DEF but M. NEG COP
 kè [ù pén kí-dìi-k gbààn.]
 that [RP roof 11-room-11 DEF]
 Intended: ‘Sukpen must roof the room, but Mpopinn mustn’t.’
- b. *Sukpen yè kè ù pén kí-dìi-k gbààn àmàà Mpopiin àà yè
 s. COP that RP roof 11-room-11 DEF but M. NEG COP
 [kè ù pén kí-dìi-k gbààn.]
 [that RP roof 11-room-11 DEF]
 Intended: ‘Sukpen must roof the room, but Mpopinn mustn’t.’

The examples in (32) further demonstrate that the modal adverb *nibààkàn* ‘maybe’ and the deontic modal *lì* are unable to license VP ellipsis. It does not matter if *nibààkàn* occurs clause-initially, medially or finally, the resulting sentence is ungrammatical (32b).

- (32) a. *Kojo lì dàà chééchè àmàà Fred àà lì [dàà chééchè.]
 K. should buy.PFV 3.bicycle but F. NEG should [buy bicycle]
 Intended: ‘Kojo should buy a bicycle, but Fred shouldn’t.’
- b. *Yelson gà nibààkàn gbìi bànehè àmàà Philip ààn nibààkàn
 Y. FUT maybe dig 2.cassava and P. NEG.FUT maybe
 [gbìi bànehè.]
 [dig 2.cassava]

This section explored VP ellipsis in Likpakpaanl and showed that there is a restricted set of modal auxiliaries which can license VPE, viz *bàn* and *ɣmàn*. The data further showed that both modals target their complements for deletion, although they elide different constituents. I also presented evidence that demonstrates the inability of the modals *yè kè*, *nibààkàn* and *lì* to license VP ellipsis in Likpakpaanl.

6 On the derivation of Likpakpaanl VP ellipsis

The empirical data on Likpakpaanl VP ellipsis have shown that it is licensed by a modal verb, which licenses the deletion of its complement. The data also confirm the assertion that ellipsis is a cross-linguistics phenomenon even though the licensing heads vary from one language to another. I account for the Likpakpaanl data by adopting Merchant’s (2001) ellipsis [E]-feature analysis and postulate that VP ellipsis in Likpakpaanl is triggered by an E-feature, which is

merged on the head of the modal phrase (Mod^0). According to Merchant (2001), the [E]-feature embodies semantic, syntactic, and phonological feature bundles that interact with the licensing head to trigger ellipsis. Semantically, only constituents with a salient antecedent, i.e. already mentioned in a previous discourse, e-GIVEN, in the sense of (Schwarzschild 1999, Merchant 2001), can undergo deletion. In terms of syntax, each language has certain syntactic categories that are eligible for the E-feature which licenses ellipsis. For example, only finite auxiliary verbs may bear the feature in English, while in French, it is the modal verb (Johnson 2001, Aelbrecht 2010, Dagnac 2010). Merchant (2001) further argues that the phonological component of the [E]-feature is responsible for removing the PF content of the complement of the head bearing the feature.

From the data presented in this paper, I contend that Likpakpaanl modals possess the [E]-feature for licensing VP ellipsis. Syntactically, the [E]-feature in Likpakpaanl is located on the head of the modal phrase (Mod^0) and targets the complement to the modal for deletion. Once the [E]-feature is activated on (Mod^0), it triggers the “deletion” of the PF content of its complement if it has an appropriate salient antecedent. After the ellipsis, the E-site becomes inaccessible for any further syntactic operations and Merchant (2004) uses that as a basis to explain why some island violations are obviated under ellipsis. Thus, I assume that the sentence in (33) has the underlying syntactic structure represented in Figure 1.⁵

- (33) Nantob àà nànn ɲmàn dàà lòòr pùàn gbààn àmàà Mpopiin gà ɲmàn
 N. NEG PST can buy 3.car new DEF but M. FUT can
 [dàà lòòr pùàn gbààn.]
 [buy 3.car new DEF]
 ‘Nantob was not able to buy the new car, but Mpopiin will be able to [buy
 the new car].’

I assume, following Munn (1987), that the conjunction *àmàà* is a functional head projecting a Boolean Phrase (BP). The first conjunct in the example in (33) occurs in the specifier to the Boolean head while the complement of the B-head, in which VP ellipsis takes place, is the second conjunct of the coordinated constituent (Munn 1992). To derive the structure in Figure 1, the VP is merged with *little v* followed by a head movement of the *dàà* from V to *little v*. The vP external argument *Mpòpíín* then moves from its base-generated position in Spec-vP to Spec-TP to satisfy the Extended Projection Principle (EPP) requirement on T. The [E]-feature on the lexical head *ɲmàn* licenses the deletion of the PF-content

⁵The marked rectangular area on the tree structure represents the ellipsis site.

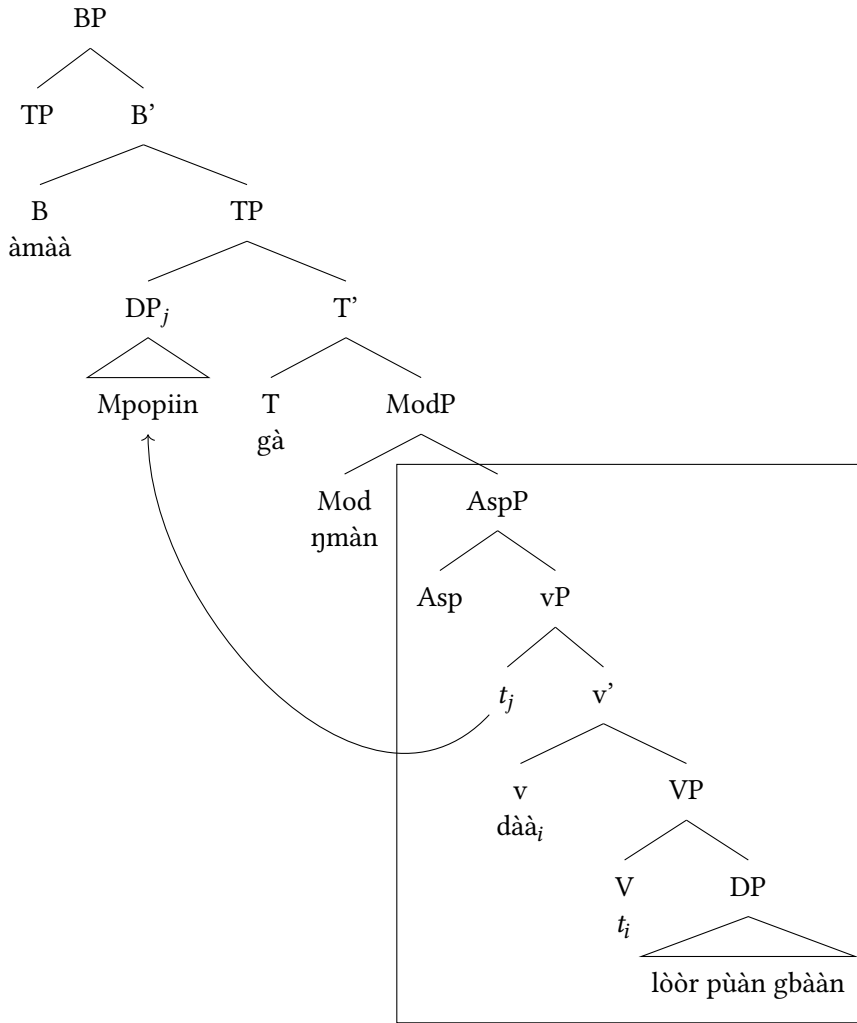


Figure 1: Syntactic tree for (33)

of vP complement and its internal arguments since they satisfy the e-GIVENness condition.

On the basis of the distribution of *bàn*, I assume that the VP ellipsis in Figure 2 is licensed by the modal head *bàn* bearing the [E]-feature that deletes the CP-complement to the (Mod⁰). The major difference between *ɲmàn* and *bàn* as shown in (34) and illustrated in Figure 2 is that the latter targets an entire complementiser phrase rather than the VP as in the case of Figure 2. It has been

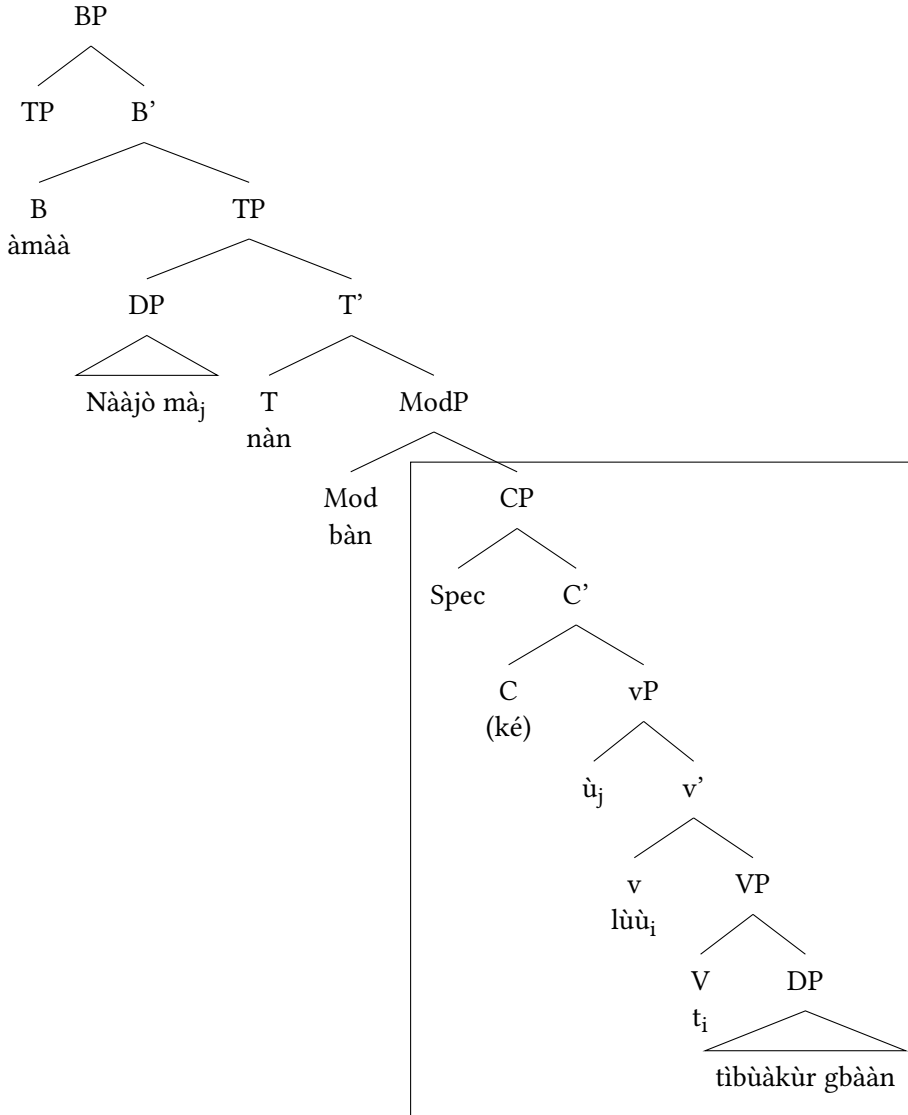


Figure 2: Syntactic tree for (34)

established already in (30) and (31) that the resumptive pronoun inside the CP is base-generated in Spec-TP and gets its interpretation from the c-commanded antecedent in Spec-TP of the matrix clause. Thus, even though the VP *...weave the two baskets* is present throughout the derivation, it is subject to deletion at PF, and the semantics provide the interpretation at LF.

- (34) Kulma àà nànn bàn (ké) ù lùù tì-bùàkùr tì-lèè gbàànn àmàà Nààjò
 K. NEG PST want that RP weave 14-basket 14-two DEF but N.
 mà nànn bàn [(ké) ù lùù tìbùàkùr gbàànn.]
 mà PST want [that RP weave 14-basket 14-two DEF]
 ‘Kulma didn’t want to weave the two baskets, but as for Naajo, she
 wanted to [weave the baskets.]’

Merchant (2019, 2001) notes that cross-linguistically, there is a restriction on syntactic elements and categories that can license ellipsis. Likpakpaanl VP ellipsis data show that the [E]-feature is syntactically restricted and optionally occurs on only modal heads that select the site. The ungrammatical sentences in (35), thus, demonstrate that lexical elements marking tense (35a), aspect (35b) and even negation (35c) are unable to license VP ellipsis in Likpakpaanl.

- (35) a. *Bààyi àànn kpiìn ì-nyòò gbàànn, àmàà Nímbé gà [kpiìn ì-nyòò.]
 B. NEG.FUT feed 6-goat DEF but N. FUT [feed 6-goat]
 Intended: ‘Baayi will not feed the goats, but Nimbe will.’
 b. *Pònjà bì kòr ì-gbéér àmàà Wùmbèi àà bì [kòr
 P. IPFV slaughter 6-pig but W. NEG IPFV [slaughter
 ì-gbéér.]
 6-pig]
 Intended: ‘Pònjà is slaughtering pigs, but Wùmbèi isn’t.’
 c. *Nààjòì bàn ké ùì chààr ì-mùùl gbàànn, bèè ù àà [bàn ké
 N. want that RP winnow 10-rice DEF or 3SG NEG [want that
 ùì chààr ì-mùùl gbàànn.]
 3SG winnow 10-rice DEF]
 Intended: ‘Does Nààjò want to winnow rice or she doesn’t?’

The discussion so far has shown that both *nyàn* and *bàn* can license VP ellipsis, but one theoretical question that needs to be answered is whether each of these modals possesses the [E]-feature and if so, how is VP ellipsis licensed when both modals co-occur in a sentence as (36a). (36a) shows that when the two modals co-occur, it is the hierarchically higher root modal *nyàn* that can license the ellipsis of its complement and not the lower modal *bàn* as reflected in the ungrammatical sentence in (36b).

- (36) a. Kùnjì gà ñmàn bàn_i ù_i dàà lòòr àmàà ù àà kpà í-lík
 K. FUT can want 3SG_i buy 2.car but 3SG_i NEG have 6-money
 nìmàpùnà, ù àà ñmàn [bàn ù dàà lòòr.]
 so, 3SG_i NEG can [want RP buy 2.car.]
 ‘Kùnjì would want to buy a car, but he has no money, so he can’t.’
 b. *Kùnjì gà ñmàn bàn ù dàà lòòr àmàà ù àà kpà
 K. FUT can.MOD want 3SG_i buy 2.car but 3SG_i NEG have
 í-lík nìmàpùnà, ù àà ñmàn bàn [ù dàà lòòr.]
 6-money so, 3SG_i NEG can want [RP buy 2.car.]
 Intended ‘Kùnjì would want to buy a car, but he has no money, so he
 can’t.’

The type of VP ellipsis manifested in Likpakpaanl is similar to what obtains in French, Italian, and Dutch. The examples in (37a) and (37b) show that VP ellipsis in French and Spanish, respectively, are licensed by modals.

- (37) a. French (Dagnac 2010: 158)
 Tom a pu voir Lee, mais Marie n’a pas pu.
 T. can.PST see L. but M. not can.PST
 ‘Tom could see Lee, but Mary couldn’t.’
 b. Spanish (Dagnac 2010: 158)
 Tom pudo ver a Lee, pero María no pudo.
 T. can.PST see (to) L. but M. not can.PST
 ‘Tom could see Lee, but Mary couldn’t.’

A similar observation has been made by Aelbrecht (2010) showing that in Dutch and Italian, too, only root modal verbs can license VP ellipsis as illustrated in (38a) and (38b) for Dutch and Italian, respectively.

- (38) a. Dutch (Aelbrecht 2010: 14)
 Jelle zal wel betalen, maar Johan kan niet.
 J. will PRT pay but J. can not
 ‘Jelle will pay, but Johan can’t.’
 b. Italian (Aelbrecht 2010: 14)
 Antonio pagherà, ma Stefano non può.
 A. will.pay but S. not can
 ‘Antonio will pay, but Stefano can’t.’

The argument advanced in this paper is that the syntactic position where the [E]-feature can be hosted to license VP ellipsis is in the position occupied by *ñmàn*.

As a result, *bàn* may only license ellipsis in the absence of *ɲmàn*. The syntactic distribution of *bàn* is similar to Dutch root modal verbs, which, according to Aelbrecht (2010), select a TP complement and not a VP. He also notes that when the two modals co-occur as in (39a), it is the hierarchically high modal *moet* and not *kunnen* that licenses ellipsis. Unlike in Likpakpaanl where the modal *ɲmàn* occurs higher and licenses ellipsis when it co-occurs with *bàn*, in Dutch, it is the opposite; the root modal *kunnen* which expresses “ability” does not license ellipsis as shown in the ungrammatical sentence in (39b).

(39) Dutch (Aelbrecht 2010: 151)

- a. Hij zal geen Frans kunnen spreken, maar ik vind dat hij eigenlijk
he will no French be.able speak but I find that he actually
wel moet [~~? Frans spreken~~ / ? Frans kunnen spreken.]
PRT must [French speak French be.able speak]
‘He won’t be able to speak French, but I think that he actually should
(speak French/be able to speak French).’
- b. *Hij zal geen Frans kunnen spreken, maar ik vind dat hij eigenlijk
he will no French be.able speak but I find that he actually
wel moet kunnen [~~Frans spreken.~~]
PRT must be.able [French speak]

Thus, while VPE in English can be licensed by a finite auxiliary (40a), a modal (40b), or the infinitival marker *to* (40c) as a licenser for VP ellipsis, it is not the same in Likpakpaanl and Dutch.

- (40) a. Mr HoneyPie was happily eating pumpkins, but his wife certainly
wasn’t.
- b. Laura was hit by the elephant trunk, and Lola was [~~hit by the
elephant trunk~~], too.
- c. Bettina wants to hear Susanna’s story, and I also want to.

(Aelbrecht & Harwood 2019: 506, 512)

Johnson (2001) and van Craenenbroeck (2017) note that in English, only finite auxiliaries can license VP ellipsis even though there are instances such as (41) in English where the sentential negator, *not*, can sometimes license VP ellipsis.

- (41) John is leaving but Mary’s not [~~leaving~~]. (Lobeck 1995: 143)

The behaviour observed with the modal auxiliary *bàn* is not unique to Likpakpaanl because previous research on VP ellipsis has shown that remnant elements,

including aspectual and voice heads, can intervene between the modal and the elided verb. As Merchant (2013) explains, in (42), the finite auxiliary *should* elides the vP, but the non-finite auxiliary intervenes between the licensing head and the ellipsis site.

- (42) The janitor must remove the trash whenever it is apparent that it should be removed. (Merchant 2013: 78)

We have illustrated VP ellipsis in Likpakpaanl and demonstrated that such ellipsis is licensed by root modal auxiliaries, which elide the phonological content of their complements. Furthermore, I have argued that the ellipsis site possesses a complex syntactic structure that is absent from the LF. In Section 6.1, I adduce syntactic evidence for a PF-deletion account for Likpakpaanl VP ellipsis.

6.1 Motivating a PF-deletion approach to Likpakpaanl VP ellipsis

This section presents evidence supporting a PF-deletion analysis for VP ellipsis in Likpakpaanl. Under the PF-deletion approach, the ellipsis site is argued to have a full-fledged structure. Therefore, there are no semantic or syntactic differences between elliptical and non-elliptical sentences, except that the [E]-feature deletes the PF structure in the former. This approach to ellipsis postulates that the ellipsis site has a fully-fledged syntactic structure and therefore has the same underlying syntax as their non-ellipsis elided counterparts (Ross 1969, Johnson 2001, Merchant 2001, 2019).

Drawing on examples from Antecedent-Contained Deletion (ACD) and island constraints, I argue for the presence of structure at the ellipsis site in Likpakpaanl VP ellipsis. Antecedent-contained ellipsis describes a phenomenon where an elided verb phrase is contained within its own antecedent. For instance, in (43a), the verb phrase in the main clause appears to license ellipsis inside the relative clause, which modifies its object. The VP *buy all the clothes* has a salient antecedent (e-GIVEN) in the first clause and is elided inside the relative clause. The deleted VP at the ellipsis site can only mean *buy all the clothes*, indicating that the object, though raised out of the VP, is still interpreted in a VP-internal position. (43b) still exemplifies the same ACD except that in this case, the ellipsis site targeted by the modal *bàn* is the entire CP and not just the verb phrase and its arguments. The presence of such ACD constructions, which require the DP to be moved from a base position prior to ellipsis, supports the assertion that the ellipsis site has a fully-fledged syntactic structure that is unpronounced at LF.

- (43) a. Kunjam nànn dàà ò-*kèkèn* ò-*mòmòk*_i ù gà òmàn [dàà t_i] nà.
 K. PST buy 8-cloth 8-all 3SG FUT can [dàà] REL.DET
 ‘Kunjam bought all the clothes that he could.’
 b. Obed nànn bàn ké ù kàrn tì-gbàn tì-*mòmòk*_i Alex nànn bàn [ké
 O. PST want that RP read 14-book 14-all A. PST want [that
 ù kàrn t_i] nà.
 RP read] REL.DET
 ‘Obed wanted to read every book that Alex wanted.’

Additional A-bar movement examples (44a) demonstrate that it is possible to extract an object out of the verb phrase prior to the ellipsis and leave a trace in the base generated position. I assume that A-bar moved constituents would not have a position (trace) in their base-generated position if the ellipsis site did not contain any underlying syntactic structure (cf: Merchant 2001, 2004, van Craenenbroeck 2017, Lipták & Aboh 2013). As predicted, when the DP is inside an island as in (44b), it is impossible for extraction to take place because it violates the complex NP island constraint, which prohibits movement out of the island.

- (44) a. Philemon nyì ò-kòlà òj Líyàb_i nànn òmàn kòr nà,
 P. know 6-fowl 6.REL L. PST can slaughter REL.DET
 àmàà ù àà nyì ò-kòlà òj ù_i àà nànn òmàn [kòr t_j]
 but 3SG NEG know 6-fowl 6.REL 3SG_i NEG PST can [slaughter]
 nà.
 REL.DET
 ‘Philemon knows the fowls that Liyab was able to slaughter, but he doesn’t know which ones he couldn’t.’
 b. *Nyoja nànn òmàn tər bì-kpààb bì kpà ò-nà-jà_i nà, àmàà
 N. PST can help 2-farmer 6.REL have 6-cow-man REL.DET but
 m àà nyì [DP ò-nà-jà òjà] ù nànn òmàn [tər
 1SG. NEG know 6-cow-man how.many 3SG PST can [help
 bì-kpààb bì kpà t_{DP}] nà.
 2-farmer 6.rel have] REL.DET
 Intended: ‘Nyoja was able to help the farmers who had bulls, but I don’t know how many bulls he could.’

In this sub-section, I have attempted to provide some formal evidence to support the argument that there is an unpronounced underlying syntactic structure at the ellipsis site in Likpakpaanl VP ellipsis.

7 Conclusion

This paper provided an account of the derivation of verb phrase ellipsis in Likpakpaanl and showed that the language exhibits a type of VP ellipsis where the complement to the modal auxiliaries is targeted for deletion. Drawing on empirical data, I showed that VP ellipsis is licensed by a restricted class of modals, namely, *ɲmàn* and *bàn*, which functionally project a Modal phrase (ModP) below T and above AspP and vP. In light of the current ellipsis theory of Merchant (2001) and Aelbrecht (2010), I assume that it is the [E]-feature on *ɲmàn* and *bàn* that licenses the deletion of the PF content of the complement to the licensing head (Mod^o). I further adduced evidence from ACD and complex noun phrase islands to support the assumption that there is an underlying syntactic structure at the ellipsis derivation of VP ellipsis in Likpakpaanl. The significant contribution of this paper to the understanding of ellipsis is that it has provided novel data as further evidence in support of Modal Complement Ellipsis (MCE), which has been shown to exist in French, Italian and Spanish. Aside from the empirical material that contributes significantly to the literature on VP ellipsis in Mbia languages, formal accounts have been provided for the descriptive facts on Likpakpaanl VP ellipsis.

Abbreviations

1	first person	NEG	negative
2	second person	NEG.FUT	negative future
3	third person	PFV	perfective
COP	copular	PL	plural
DEF	definite	PRT	particle
FUT	future	PST	past tense
FOC	focus	REL.DET	relative determiner
HEST.PST	hesternal past	REL	relativiser
HOD.PST	hodiernal past	RP	resumptive pronoun
IPFV	imperfective	SG	singular
LOC	locative	VENT	ventive
MOD	modal		

Acknowledgements

I wish to thank the reviewer(s) for their insightful comments and suggestions that helped me to fine-tune the ideas in this paper. I am also grateful to my col-

leagues from the Syntax reading group at Goethe University, Frankfurt am Main, and Johannes Mursell and Anke Himmelreich for their suggestions and positive comments. I wish to thank the participants at the Syntax Colloquium, Goethe University for their input and comments that helped to shape this work. I also thank the organisers of the African Linguistics School (ALS 6) for providing me with the platform to present this work and for the wonderful feedback that I was given.

References

- Abubakari, Hasiyatu. 2018. *Aspects of Kusaal grammar: The syntax-information structure interface*. University of Wien. (Doctoral dissertation).
- Abubakari, Hasiyatu. 2019. The syntax and semantics of relative clauses in Kusaal. *Ghana Journal of Linguistics* 8(2). 27–62.
- Aelbrecht, Lobke. 2010. *The syntactic licensing of ellipsis*. Amsterdam: John Benjamins Publishing Company.
- Aelbrecht, Lobke & William Harwood. 2015. To be or not to be elided: VP ellipsis revisited. *Lingua* 153. 66–97.
- Aelbrecht, Lobke & William Harwood. 2019. Predicate ellipsis. In Jeroen van Craenenbroeck & Tina Temmerman (eds.), *The Oxford handbook of ellipsis*, 504–525. Oxford: Oxford University Press.
- Ameka, Felix K. 2008. Aspect and modality in Ewe: A survey. In Felix K. Ameka & Mary Esther Kropp Dakubu (eds.), *Aspect and modality in Kwa languages*, 135–194. Amsterdam: John Benjamins Publishing Company.
- Atintono, Samuel Awinkene. 2013. *The semantics and grammar of positional verbs in Gurene: A typological perspective*. University of Manchester. (Doctoral dissertation).
- Bassong, Paul Roger. 2014. *Information structure and the Basa'a left peripheral syntax*. University of Yaoundé, Cameroon. (Doctoral dissertation).
- Bilola, Edmond & Paul R. Bassong. 2020. Sluicing and functional heads in Bantu. In Edmond Bilola (ed.), *Cartography and antisymmetry: Evidence from Bantu and Chadic*, 218–237. Munich: Lincom Europa.
- Bisilki, Abraham Kwesi & Rebecca Atchoi Akpanglo-Nartey. 2017. Noun pluralisation as a dialect marker in Likpakpaln-‘Konkomba’. *Journal of West African Languages* 44(2). 24–42.
- Bodomo, Adams. 1997. *The structure of Dagaare* (Stanford Monographs in African Languages). Stanford: CSLI Publications.

- Bodomo, Adams. 2020. Mabilia: Its etymological genesis, geographical spread and some salient genetic features. In *Handbook of the Mabilia languages of West Africa*. Adams Bodomo, Hasiyatu Abubakari & Samuel A Issah (eds.). Glienicke: Galda Verlag. 5–34.
- Butler, Jonny. 2003. A minimalist treatment of modality. *Lingua* 113(10). 967–996. DOI: 10.1016/S0024-3841(02)00146-8.
- Chomsky, Noam. 1980. On binding. *Linguistic Inquiry* 11. 1–46.
- Chomsky, Noam. 1981. *Lectures on government and binding*. Dordrecht: Foris.
- Chomsky, Noam. 2000. Minimalist inquiries: The framework. In Martin Roger, Michaels David, Uriagereka Juan & Jay Keyser Samuel (eds.), *Step by step. Essays on minimalist syntax in honor of Howard Lasnik*, 89–155. Cambridge: Cambridge, MA: The MIT press.
- Cinque, Guglielmo. 1999. *Adverbs and functional heads: A cross-linguistic perspective*. Oxford: Oxford University Press.
- Dagnac, Anne. 2010. Modal ellipsis in French, Spanish and Italian: Evidence for a TP-deletion analysis. In Karlos Arregi, Zsuzsanna Fagyal, Silvina A. Montrul & Aafke Tremblay (eds.), *Romance linguistics 2008: Interactions in Romance*, 157–170. Amsterdam: John Benjamins Publishing Company.
- Dalrymple, Mary, Stuart M. Shieber & Fernando CN Pereira. 1991. Ellipsis and higher-order unification. *Linguistics and Philosophy* 14. 399–452.
- Fiengo, Robert & Robert May. 1994. *Indices and identity*. Cambridge: The MIT Press.
- Frazier, Lyn & Charles Clifton. 2005. The syntax-discourse divide: Processing elliptical constructions. *Syntax* 8(2). 121–174.
- Giorgi, Alessandra. 2013. Epistemic adverbs, the prosody-syntax interface and the theory of phases. In Christina Tortora, Marcel den Dikken, Ignacio L. Montoya & Teresa O'Neill (eds.), *Romance Linguistics 2013: Selected Papers from the 43rd Linguistic Symposium on Romance languages*, 99–118. New York: John Benjamins Publishing Company.
- Hacquard, Valentine. 2006. *Aspects of modality*. Massachusetts Institute of Technology. (Doctoral dissertation).
- Hacquard, Valentine. 2009. On the interaction of aspect and modal auxiliaries. *Linguistics and Philosophy* 32(3). 279–315. DOI: 10.1007/s10988-009-9061-6.
- Hacquard, Valentine. 2010. On the event relativity of modal auxiliaries. *Natural Language Semantics* 18(1). 79–114. DOI: 10.1007/s11050-010-9056-4.
- Hacquard, Valentine. 2016. Modals: Meaning categories. In Joanna Blaszczak, Anastasia Giannakidou, Dorota Klimek-Jankowska & Krzysztof Migdalski (eds.), *Mood, aspect, modality revisited: New answers to old questions*, 45–74. Chicago: The University of Chicago Press.

- Hardt, Daniel. 1999. Dynamic interpretation of verb phrase ellipsis. *Linguistics and Philosophy* 22(2). 187–221.
- Heine, Bernd. 1991. Auxiliaries in African languages: The Lingala case. In Kathleen Hubbard (ed.), *Proceedings of the seventeenth Annual Meeting of the Berkeley Linguistics Society: Special session on African language structures*, vol. 17, 106–119.
- Hoye, Lars. 1997. *Adverbs and modality in English*. London: Longman.
- Issah, Samuel Alhassan. 2013. The function of the post verbal *la* in Dagbani. *Studies in African Linguistics* 42(2). 153–74.
- Issah, Samuel Alhassan. 2020. *On the structure of A-bar constructions in Dagbani: Perspectives of wh-questions and fragment answers*. Switzerland: Peter Lang.
- Johnson, Kyle. 2001. What VP ellipsis can do, and what it can't, but not why. In Mark Baltin & Chris Collins (eds.), *The handbook of contemporary syntactic theory*, 439–479. Oxford: Blackwell Publishers.
- Joshua Project. 2022. *Konkomba people group in all countries*. https://joshuaproject.net/people_groups/10842 (14 February, 2023).
- Kratzer, Angelika. 1981. The notional category of modality. In Hans-Jürgen Eikmeyer & Hannes Rieser (eds.), *Words, worlds, and contexts. New approaches in word semantics*, 289–323. Berlin: De Gruyter Mouton.
- Kropp Dakubu, Mary Esther. 2003. Aspect agreement constraints in Gurene multi-verb constructions. In Dorothee Beermann & Hellan Lars (eds.), *Multi-verb constructions*. Trondheim: Trondheim Summer School.
- Lipták, Anikó & Enoch O. Aboh. 2013. Sluicing inside relatives: The case of Gungbe. *Linguistics in the Netherlands* 30(1). 102–118.
- Lobeck, Anne. 1995. *Ellipsis: Functional heads, licensing, and identification*. Oxford: Oxford University Press.
- Manessy, Gabriel. 1971. Les langues gurma. *Bulletin de l'Institut Fondamental d'Afrique Noire, Série B: Sciences humaines* 33(1). 117–246.
- Mendes, Gesoel & Jason Kandybowicz. 2023. Salvation by deletion in Nupe. *Linguistic Inquiry* 54(2). 299–325. DOI: 10.1162/ling_a_00434.
- Merchant, Jason. 2001. *The syntax of silence: Sluicing, islands, and the theory of ellipsis*. Oxford: Oxford University Press.
- Merchant, Jason. 2004. Fragments and ellipsis. *Linguistics and Philosophy* 27. 661–738. DOI: 10.1007/s10988-005-4459-9.
- Merchant, Jason. 2013. Voice and ellipsis. *Linguistic Inquiry* 44(1). 77–108.
- Merchant, Jason. 2019. Ellipsis: A survey of analytical approaches. In Jeroen van Craenenbroeck & Tina Temmerman (eds.), *Oxford handbook of ellipsis*, 19–45. Oxford: Oxford University Press.

- Munn, Alan. 1987. Coordinate structures, x-bar theory and parasitic gaps. *McGill Working Papers in Linguistics* 4(1). 121–140.
- Munn, Alan. 1992. A null operator analysis of ATB gaps. *The Linguistic Review* 9(1). 1–26.
- Naden, Tony. 1989. Gur. In John Bendor-Samuel & Rhonda L. Hartell (eds.), *The Niger-Congo languages: A classification and description of Africa's largest language family*, 140–168. Lanham: University Press of America.
- Nuyts, Jan. 1993. Epistemic modal adverbs and adjectives and the layered representation of conceptual and linguistic structure. *Linguistics* 31(5). 933–970. DOI: 10.1515/ling.1993.31.5.933.
- Nuyts, Jan. 2001. *Epistemic modality, language, and conceptualization*. Amsterdam: John Benjamins Publishing Company.
- Reinhart, Tanya & Eric Reuland. 1993. Reflexivity. *Linguistic Inquiry* 24(4). 657–720.
- Ross, John Robert. 1967. *Constraints on variables in syntax*. Cambridge: Massachusetts Institute of Technology. (Doctoral dissertation).
- Ross, John Robert. 1969. Guess who? In *Papers from the 5th Regional Meeting of the Chicago Linguistic Society*, 252–286. Chicago.
- Schwarzschild, Roger. 1999. Givenness, AvoidF and other constraints on the placement of accent. *Natural Language Semantics* 7(2). 141–177.
- Tait, David & Peter D. Strevens. 1954. Konkomba nominal classes. *Africa* 24(2). 130–148.
- Uchechukwu, Chinedu. 2008. The modal system of the Igbo language. In Werner Abraham & Elisabeth Leiss (eds.), *Modality-aspect interfaces: Implications and typological solutions*, 241–278. Amsterdam: John Benjamins Publishing Company.
- van Craenenbroeck, Jeroen. 2017. VP-ellipsis. In Martin Everaert & Henk C. van Riemsdijk (eds.), *The Wiley Blackwell companion to syntax*, 2nd edn., 1–35. Hoboken: John Wiley & Sons Inc.
- Winkelmann, Kerstin. 2012. D4. Konkomba (Likpakpaln). In Gudrun Mieke, Brigitte Reineke & Kerstin Winkelmann (eds.), *Noun class systems in Gur languages: North Central Gur languages*, vol. 4, 472–486. Cologne: Köppe.
- World Population Review. 2023. *Ghana Population 2023: Ghana Demographics*. <https://worldpopulationreview.com/countries/ghana-population> (13 February, 2023).

